AI AND VISUALIZATION 2026W · PHASE I MIDPOINT

The 20 Hz Club

Where and why do three whale call detectors fail?

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Phase I: how does the model behave?
Three call classes: blue whale calls (BMABZ), D-calls (D), fin whale pulses (BP)

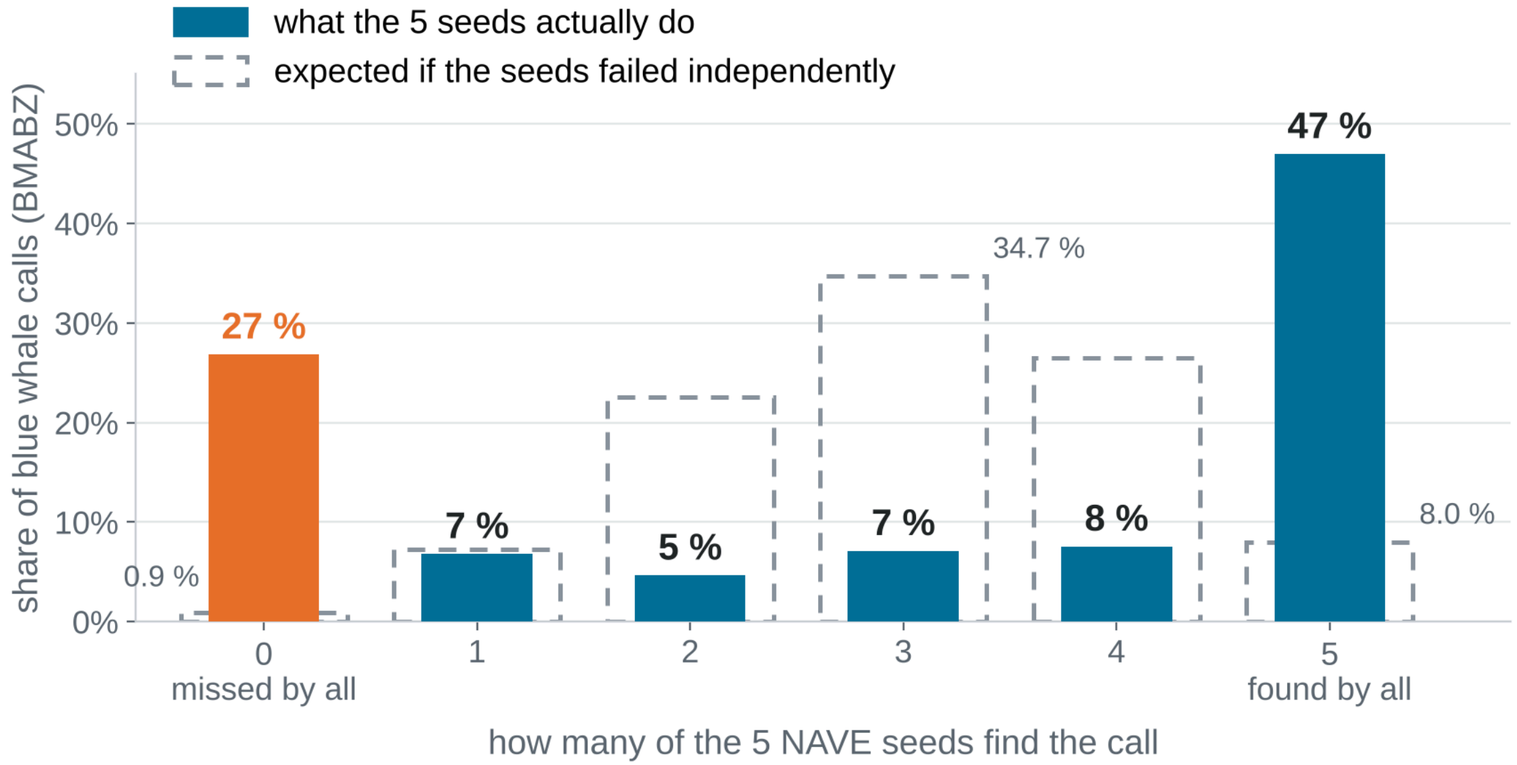


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[1] BioDCASE 2026 Task 2 baseline system, Ultralytics YOLO11, biodcase.github.io. [2] C. M. Geldenhuys, G. Tonitz, T. R. Niesler, “Whale-VAD: Whale vocalisation activity detection,” DCASE 2025 Challenge, tech. rep., 2025. [3] M. Nagl, P. Primus, G. Widmer, “NAVE: A normalized, adaptive Conformer for Antarctic whale vocalization detection,” DCASE 2026 Workshop, 2026.

PHASE I · SAME MISSES Human-only pair

NAVE seeds miss the same blue whale calls



27 %
of blue whale calls are missed by all five seeds

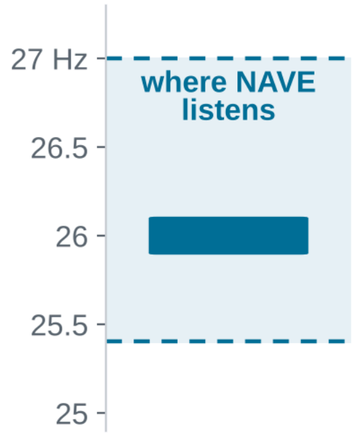
0.9 %
expected if the five seeds failed independently

47 %
are found by all five: a call is easy or impossible

PHASE I · PITCH Human-only pair numbers: AI-aided replication

The models only hear the training pitch

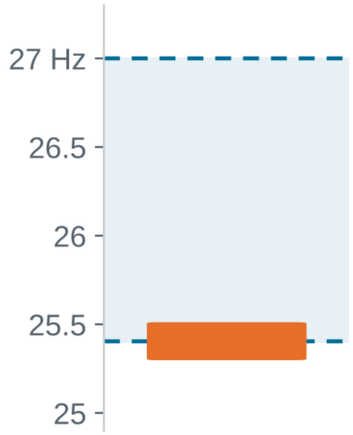
1 Training pitch



55

of 100 known blue whale calls found

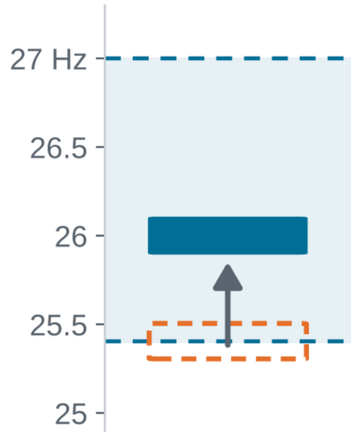
2 2019/20 pitch



19

of 100 known blue whale calls found

3 2019/20 audio shifted up



×2.5

NAVE blue whale calls per hour: 7.2 → 18.3

×2.5 | ×0.01

NAVE blue whale calls per hour on 2019/20 audio, up | down

×2.9 | ×0.07

the same for YOLO

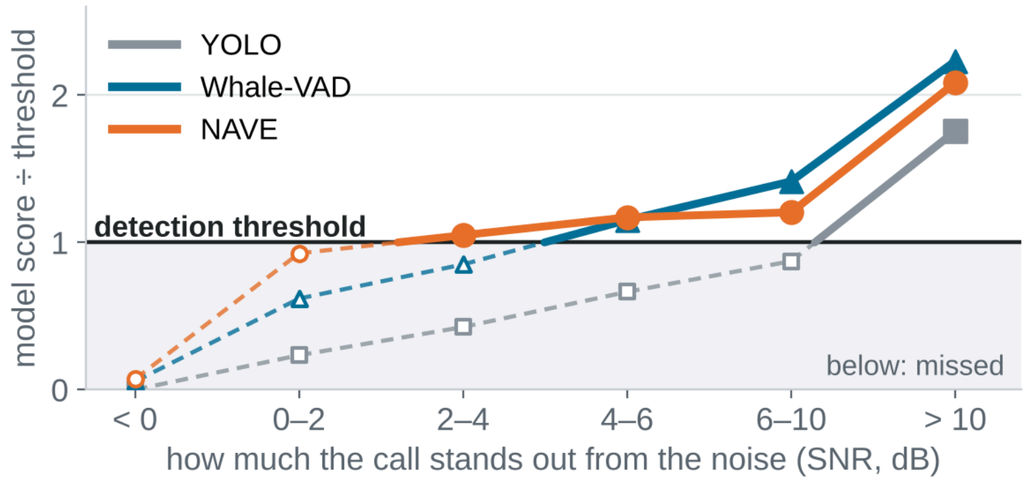
×5.3 | ×0.28

the same for Whale-VAD

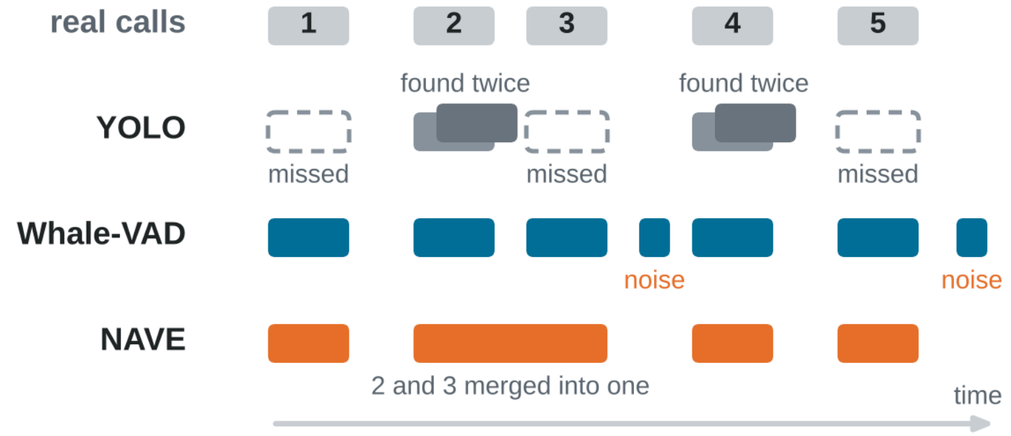
PHASE I · WHERE THEY FAIL AI-aided pair

Faint calls are missed by all three models

Faint calls never reach the threshold



How each model goes wrong (sketch)



29 %
of all calls (3 classes) missed by all 7 models

48 %
YOLO: calls with no response

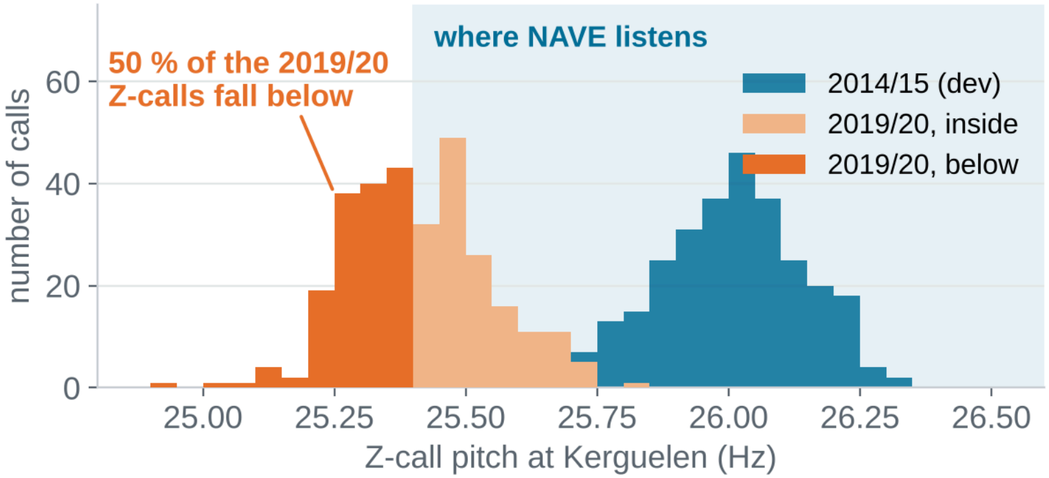
56 %
Whale-VAD: detections on noise

9 %
NAVE: calls merged with a neighbour

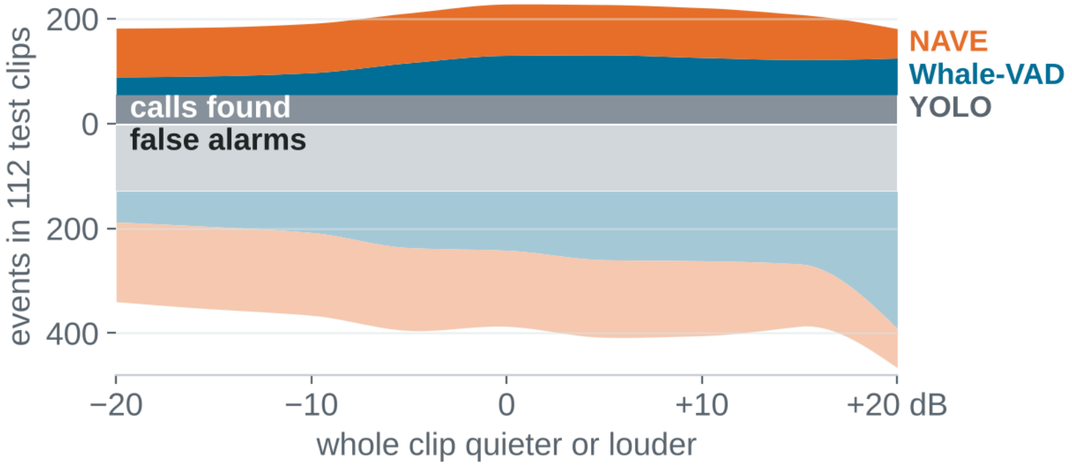
PHASE I · WHY THEY FAIL AI-aided pair

The models hear only a narrow pitch band

Z-calls inside and outside the window



Loudness changes what each model finds, all classes



−0.13 Hz / year
Z-call pitch at Kerguelen, 2005 to 2020

50 %
of the 2019/20 Z-calls fall below NAVE's listening range

3 reactions
to loudness: NAVE goes quiet, Whale-VAD noisy, YOLO none